

# A multi-architecture study of specificity refinement and false-positive mechanism analysis in prostate MRI

Yongbo Shu<sup>1,2,3,4</sup>, Kewen Chen<sup>2</sup>, Yifeng Yuan<sup>1,3,4</sup>, Zirui Xin<sup>1,3,4</sup>, Luo Lei<sup>1,3,4</sup>, Yang Yang<sup>1,2,3,4</sup>, Xi Chen<sup>1,3,4,\*</sup>, Aijing Luo<sup>1,2,3,4,\*</sup>

- (1) The Second Xiangya Hospital of Central South University, Changsha, Hunan 410011, China
- (2) School of Life Sciences, Central South University, Changsha, Hunan 410013, China
- (3) Hunan Provincial Key Laboratory of Medical Information Research (Central South University), Changsha, Hunan 410011, China
- (4) Hunan Provincial Clinical Medical Research Center for Cardiovascular Intelligent Medicine, Changsha, Hunan 410011, China

\* Corresponding authors: Xi Chen, nancy\_chen@csu.edu.cn; Aijing Luo, luoj@csu.edu.cn

## Abstract

**Objectives.** To characterize residual false positives in prostate MRI detection, and to evaluate a lightweight post-hoc refinement head for case-level specificity.

**Materials and Methods.** This retrospective study used PI-CAI (5-fold cross-validation) and Prostate158 ( $n = 158$ ; external). A context-aware evidence head and an 89,216-parameter refinement head were trained on a frozen detection backbone; the evidence head was also trained on four further backbones (bare nnU-Net, bare U-Net, bare Mamba, MIGF-Mamba). For each false-positive region, T2-weighted, apparent-diffusion-coefficient, and high-b-value contrast ratios versus peri-lesional rings were compared against ground-truth lesions and contralateral benign regions.

**Results.** False positives were closer to true cancers than to benign tissue in evidence and raw T2-weighted and apparent-diffusion-coefficient contrast, reproducing 35/35 across five architectures (Cohen’s  $d$  1.10; FP/benign evidence ratio 2.38x) and 105/105 across modality-perturbation scenarios. On PI-CAI fold-0, refinement raised case-level specificity from  $0.469 \pm 0.181$  to  $0.549 \pm 0.132$  (+17.2%) at preserved sensitivity ( $0.943 \pm 0.021$ ); 5-fold cross-validation showed fold-conditional behavior (9/15 observations positive; range -22% to +28%). On Prostate158, both models saturated (McNemar pooled  $p = 0.69$ ), while the false-positive contrast-matching finding replicated.

**Conclusion.** Residual false positives are contrast-matched to cancer (sharing raw imaging features rather than histologically confirmed mimicry), reproducing across five architectures – a data-level imaging property, not model-specific artifacts; post-hoc refinement adds practical specificity in-domain but is fold-conditional.

**Keywords:** Prostate cancer; Magnetic resonance imaging; Deep learning; False positive reactions; Image interpretation, computer-assisted

## Key Points

- **Question.** Why do false positives persist in prostate MRI AI, and can a lightweight post-hoc head improve case-level specificity without sensitivity loss?
- **Findings.** False positives resembled true cancers more than benign tissue (35/35 across five architectures); a post-hoc head improved fold-0 specificity by 17.2%.

## Clinical Relevance Statement

False positives from prostate MRI AI detectors are radiologically interpretable as cancer-resembling regions, not arbitrary model artifacts; this finding reproduces across five distinct architectures and informs how AI-flagged regions should be adjudicated.

## Introduction

Clinically significant prostate cancer (csPCa) is diagnosed from multiparametric magnetic resonance imaging (MRI) followed by targeted biopsy of suspicious lesions. Prostate MRI-based artificial intelligence (AI) systems have advanced substantially on benchmark datasets such as PI-CAI [1], yet clinical translation is held back by high false-positive rates: regions flagged by AI often do not correspond to biopsy-confirmed cancer, triggering unnecessary follow-up imaging and biopsies [2, 3]. Existing AI pipelines are overwhelmingly tuned to maximize sensitivity, while specificity receives comparatively little attention and the nature of residual false positives is rarely characterized in a way that informs clinical action on AI output.

Two gaps motivate this work. First, improving case-level specificity of a deployed AI detection model typically requires full retraining; simple post-hoc recalibration (temperature or Platt scaling) cannot alter case-level specificity beyond what threshold tuning achieves, since these methods preserve ranking. Lightweight post-hoc specificity refinement on a frozen backbone that preserves sensitivity is therefore under-explored in prostate MRI. Second, false positives are usually summarized as a single error rate; their evidence structure — whether they resemble true cancers in raw imaging contrast or are arbitrary model artifacts — is rarely disentangled. Without that distinction, recommendations for radiologist adjudication and triage remain speculative.

Here we first quantify the evidence structure of residual false positives. Using a context-aware evidence head and direct raw-imaging contrast on T2-weighted and apparent-diffusion-coefficient channels, false positives from a frozen csPCa detection backbone [4] are consistently closer to true cancers than to benign prostatic tissue — a direction reproducing across PI-CAI 5-fold cross-validation, seven modality-perturbation scenarios, and five distinct backbone architectures. As an implementation this mechanism analysis explains, we introduce a lightweight (89 K-parameter) post-hoc specificity refinement head; on PI-CAI fold-0 it increases case-level specificity by 17.2% at preserved sensitivity, with 5-fold cross-validation revealing fold-conditional magnitude. We replicate both analyses on Prostate158 [5] and report what works, where center-specific effects limit the claim, and what this implies for AI-assisted prostate MRI.

# Materials and Methods

This study analyzed de-identified multiparametric prostate MRI from two publicly available, institutionally approved cohorts (PI-CAI and Prostate158) under their respective data-use licenses; no additional institutional review board approval was required and the requirement for informed consent was waived.

## 3.1 Datasets

This retrospective study used two public prostate multiparametric MRI datasets. The PI-CAI Public Training and Development Dataset [1] (1500 multi-vendor studies from 1476 patients across three Dutch centres; 425 biopsy-confirmed csPCa, 1075 pathology-negative) served as the primary cohort; we used the official 5-fold split, restricting method development and primary evaluation to fold-0 (1200 training / 300 validation). Prostate158 [5] (158 studies with biopsy-confirmed labels) served as the external cohort. Both datasets provide axial T2-weighted, high-b-value diffusion-weighted, and apparent-diffusion-coefficient sequences. Preprocessing — axial resampling to  $0.5 \times 0.5 \times 3.0$  mm, gland-centered cropping to  $128 \times 128 \times 32$  voxels, and per-channel normalization — followed our prior work [4] and was identical across datasets.

## 3.2 Frozen detection backbone

All experiments used a frozen multi-modal csPCa detection backbone from our earlier work [4]. The backbone accepts three co-registered channels (T2-weighted, high-b-value DWI, ADC) at  $128 \times 128 \times 32$  voxels; each modality has its own four-level encoder ( $3 \times 3 \times 3$  convolution, downsampling 2 per level). Adaptive gating fuses encoders at the lowest-resolution stage and feeds a shared three-stage decoder with dec1, dec2, dec3 outputs at 32/64/128 channels and full/half/quarter resolution; a  $1 \times 1 \times 1$  convolution returns per-voxel logits. Each input channel was zeroed with probability 0.3 per sample during backbone training; dropout was disabled at inference. We reused five-seed checkpoints (42, 123, 456, 789, 1024) for the A2 variant (gating + modality dropout, without deep supervision) unchanged.

We selected this backbone for three reasons: (i) **modality isolation** via per-modality encoders enables deterministic per-channel attribution required by the evidence-grounded analyses in §3.4; (ii) **clinical-grade efficiency** with 9.45 M parameters and 8.5 ms single-volume inference latency leaves compute headroom for downstream uncertainty and risk-stratification pipelines; (iii) **measurable-gain headroom** since the A2 variant’s 5-fold mean case-level specificity (0.46) avoids ceiling effects that would compress observable post-hoc refinement gains.

## 3.3 Post-hoc specificity refinement head (P2a)

The refinement head takes the backbone’s decoder features at three resolutions (dec1/dec2/dec3; 32/64/128 channels). The dec2 and dec3 streams are projected to 32 channels and trilinearly upsampled to dec1; the three streams are concatenated (96 channels), passed through a  $3 \times 3 \times 3$  convolution, instance normalization, and ReLU, then projected to a 2-channel per-voxel logit residual via a  $1 \times 1 \times 1$  convolution (initialized at zero; 89,216 trainable parameters). The residual is added to the backbone’s final-stage logits before softmax.

The head was trained for 100 epochs with Adam (learning rate  $1 \times 10^{-4}$ , weight decay  $1 \times 10^{-5}$ ) on PI-CAI fold-0 using focal loss ( $\gamma = 2$ ) plus a mean-squared regularizer on the residual. Only cases with at least one backbone-predicted false-positive region above 0.3 were sampled. Checkpoint selection

retained the epoch whose validation case-level sensitivity fell within  $\pm 0.01$  of the backbone’s; the same five seeds were trained with identical hyperparameters.

### 3.4 Evidence-grounded false-positive framework (P2b)

Per case, we extracted three ROI types: (a) the ground-truth lesion; (b) false-positive regions (backbone predictions above 0.5 with no ground-truth overlap, connected components  $\geq 10$  voxels); (c) a contralateral benign ROI by laterally mirroring the lesion. Each ROI was paired with a 5-mm peri-lesional ring via Euclidean distance transform. On T2-weighted, apparent-diffusion-coefficient, and high-b-value images we computed contrast ratios  $(m_{\text{ROI}} - m_{\text{peri}}) / (|m_{\text{peri}}| + \varepsilon)$ ; pairwise differences were tested with paired Wilcoxon signed-rank tests.

Three evidence-prediction variants of increasing context were trained: **mask-only** (11-channel geometric input  $\rightarrow$  MLP [128  $\rightarrow$  128]); **image-aware** (75-channel: mask-only + 64-channel dec2 ROI-average  $\rightarrow$  MLP [256  $\rightarrow$  256]); **context-aware** (203-channel: image-aware + 64-channel periring + 64-channel contralateral-benign features  $\rightarrow$  MLP [256  $\rightarrow$  256]). Each variant produced six regression outputs and a three-class csPCa-suspicion label, trained for 100 epochs with Adam (learning rate  $1 \times 10^{-4}$ ) on PI-CAI fold-0 using equally weighted mean-squared error and cross-entropy, with the same five seeds.

### 3.5 External validation on Prostate158

Two external analyses on Prostate158 [5] were specified before the external-cohort analysis. For P2a, case-level specificity of the bare and refined models was compared at the PI-CAI-calibrated threshold and at each model’s matched-sensitivity threshold (reproducing its PI-CAI sensitivity within  $\pm 0.02$ ). For P2b, the contrast-ratio pipeline (§3.4) was applied to Prostate158 false-positive regions to test whether T2-weighted and apparent-diffusion-coefficient distributions remained closer to true cancers than to benign tissue. No Prostate158-specific retraining or recalibration was applied.

#### 3.5b 5-fold cross-validation supplementary protocol

To assess the fold-conditional behavior of the post-hoc heads beyond the fold-0 development split, we re-trained P2a and P2b on 15 frozen base backbone checkpoints (5 PI-CAI folds  $\times$  3 seeds: 42, 123, 789), using the same hyperparameters as the fold-0 reference. Each (fold, seed) pair was evaluated on its fold-specific validation split, on Prostate158, and on the seven modality-perturbation scenarios (ideal + 3 missing-modality + 3 artifact). No hyperparameter retuning was performed.

#### 3.5c Backbone-agnostic mechanism verification

To test whether the P2b false-positive contrast-matching mechanism depends on the MIGF-nnUNet backbone, we trained the same context-aware evidence head — with identical architecture and hyperparameters — on four additional frozen backbones (bare nnUNet, bare U-Net, bare Mamba, MIGF-Mamba A2) across all 5 PI-CAI folds with seed 42, yielding 20 additional observations. We tested whether the lesion-vs-benign and false-positive-vs-benign evidence-direction findings reproduce.

### 3.6 Evaluation metrics

Lesion detection metrics: **PI-CAI Score** [6] (mean of case-level AUROC and lesion-level average precision), **case-level sensitivity**, **case-level specificity**, and **matched-sensitivity** case-level specificity (threshold reproducing each baseline’s case-level sensitivity within  $\pm 0.01$ ). Voxel-level: **positive Dice coefficient**. FP evidence analysis: contrast ratios (§3.4) on T2-weighted, apparent-diffusion-coefficient, and high-b-value channels with effect sizes against benign-ROI controls.

#### 3.6b Post-hoc recalibration baselines

Three monotonic post-hoc recalibration baselines — temperature scaling, Platt scaling, and threshold-sweep matched-sensitivity calibration — were evaluated; we showed in a separate analysis that all three reduce to threshold sweep at our case-level decision rule ( $\geq 10$ -voxel connected component), and we report the threshold-sweep result as a single representative baseline (Supplementary Table S-Baselines).

### 3.7 Statistical analysis

Descriptive summaries are five-seed mean  $\pm$  sample standard deviation (ddof = 1; seeds 42, 123, 456, 789, 1024). Inference uses case as the sampling unit: paired binary case-level outcomes via McNemar’s test (two-sided, continuity-corrected); 95% confidence intervals on paired case-level specificity differences from 10 000 per-case bootstrap resamples with seed-level estimates pooled. Five-seed Wilcoxon signed-rank tests are reported in Supplementary Material. All planned analyses are reported regardless of significance. The study follows CLAIM 2024 [7] (checklist in Supplementary Material). Analyses used Python 3.11 (PyTorch 2.11, MONAI 1.5, `picai_eval` 1.4, SciPy 1.11) on a single NVIDIA RTX 3090 GPU.

## Results

After applying the predefined fold-0 split, the primary evaluation set comprised 300 PI-CAI studies (84 csPCa, 216 pathology-negative) and the external replication set comprised all 158 Prostate158 studies (102 csPCa, 56 pathology-negative) (Figure 1, Table 1).

### 4.1 Cohort demographics

Median patient age was 66 years (IQR 62–71) in PI-CAI fold-0 validation and 64 years (IQR 58–71) in Prostate158; both cohorts were all-male (Table 1). All studies provided co-registered T2-weighted, high-b-value diffusion-weighted, and apparent-diffusion-coefficient sequences.

### 4.2 P2a specificity refinement on PI-CAI

At the PI-CAI-calibrated threshold, refinement increased case-level specificity from  $0.469 \pm 0.181$  to  $0.549 \pm 0.132$  (+17.2% relative; Table 2, Figure 2) without reducing case-level sensitivity ( $0.943 \pm 0.021$  for both). PI-CAI Score improved from  $0.730 \pm 0.056$  to  $0.741 \pm 0.047$ ; AUROC from  $0.864 \pm 0.047$  to  $0.878 \pm 0.041$ . Voxel-level positive Dice was essentially unchanged ( $0.487 \pm 0.009$  vs.  $0.490 \pm 0.005$ ), confirming the gain arose from case-level FP suppression. Across the seven standard modality-dropout scenarios [4], the gain held in all configurations retaining T2-weighted; missing-T2-weighted cases showed reduced gain (Supplementary Table S-5fold-Summary).

## 4.2b Competitive post-hoc baselines

At matched sensitivity on PI-CAI, the refinement head outperformed the monotonic-recalibration baseline by  $+0.038 \pm 0.028$  case-level specificity (4 of 5 seeds favoring P2a; Supplementary Table S-Baselines).

## 4.3 P2b evidence prediction on PI-CAI

The three evidence-prediction variants showed a stable ranking of overall mean absolute error on contrast-ratio targets across all five seeds (Table 3, Figure 3): context-aware  $0.353 \pm 0.005$ , image-aware  $0.409 \pm 0.003$ , mask-only  $0.460 \pm 0.001$ . Context-aware reduced error by 23.3% relative to the mask-only geometric baseline and by 13.7% relative to image-aware. Macro-averaged F1 on the 3-class suspicion label was similar across variants ( $0.617 \pm 0.017$ ,  $0.608 \pm 0.023$ ,  $0.616 \pm 0.034$ ), indicating the regression differences were not driven by coarse suspicion labels. Seed-level ranking held for every pairwise comparison.

## 4.4 FP evidence analysis on PI-CAI

Per-region raw contrast in PI-CAI false-positive regions consistently aligned with true cancers rather than with benign tissue (Table 4, Figure 4). In T2-weighted, false-positive 5-seed mean contrast ratio was  $-0.80 \pm 0.11$  versus  $-0.56 \pm 0.00$  for true cancers and  $+0.03 \pm 0.00$  for benign ROIs. In apparent-diffusion-coefficient, false positives were hypointense (mean  $-0.56 \pm 0.07$ ), matching the direction of true cancers ( $-0.31$ ) while benign ROIs were near-zero (0.00). In both channels, false-positive-to-benign separation exceeded false-positive-to-cancer separation across all five seeds. The high-b-value channel showed the same tendency (FP  $+0.47 \pm 0.45$  vs. GT  $+0.39$  vs. benign  $-0.03$ ) with substantially higher cross-seed variance (supporting observation).

### 4.4b False-positive similarity-tier stratification

Post-hoc refinement suppressed false-positive regions uniformly across cancer-similarity tertiles (high 28%, mid 26%, low 31% on PI-CAI; Supplementary Table S-Stratification), indicating a similarity-tier-independent suppression mechanism rather than a selective filter.

### 4.4c 5-fold cross-validation

Across 15 PI-CAI observations (5 folds  $\times$  3 seeds; ideal scenario), P2a mean case-level specificity was  $0.493 \pm 0.150$  (Supplementary Tables S-5fold-Summary, S-5fold-PerFold). Direction was positive in 9 of 15 observations; per-fold relative change versus the Paper 1 A2 5-fold baseline (0.4562) was +19.7%, +24.1%,  $-9.4\%$ , +27.8%,  $-22.4\%$  for folds 0–4 (folds 0/1/3 reproduce or exceed the fold-0 +17.2% reference; folds 2/4 reverse). P2b evidence MAE was  $0.344 \pm 0.014$ ; lesion-versus-benign and false-positive-versus-benign directions both reproduced 15/15, with Cohen’s d 1.10 (range 0.91–1.36) and FP/benign evidence ratio  $2.38\times$  (range  $1.91\times$ – $2.69\times$ ). Across seven robustness scenarios  $\times$  15 observations, lesion-versus-benign reproduced 105/105. The P2a quantitative effect is fold-conditional; the P2b mechanism direction is fold-invariant.

### 4.4d Backbone-agnostic mechanism verification

Training the P2b evidence head on four additional frozen backbones (bare nnUNet, bare U-Net, bare Mamba, MIGF-Mamba A2) across 5 PI-CAI folds with seed 42 produced 20 additional observations; the lesion-versus-benign direction reproduced in 20 of 20 and the false-positive-versus-benign

direction in 20 of 20 (Supplementary Table S-35backbones). Combined with the P2b 5-fold reference (15 of 15), the mechanism direction reproduced in 35 of 35 observations across five distinct architectures (bare nnUNet, bare U-Net, bare Mamba, MIGF-Mamba A2, MIGF-nnUNet A2), indicating that the false-positive contrast-matching phenomenon is a data-level imaging property of prostate MRI rather than a model-specific artifact.

#### 4.5 External validation on Prostate158

On Prostate158, both models saturated near full sensitivity at the PI-CAI-calibrated threshold (Table 5): bare  $1.000 \pm 0.000$ , refined  $0.998 \pm 0.004$ ; case-level specificity collapsed to near zero (bare  $0.000 \pm 0.000$ ; refined  $0.007 \pm 0.016$ ), consistent with an apparent-diffusion-coefficient domain shift [4]; aggregate discrimination was preserved (AUROC  $0.655 \pm 0.035$  vs.  $0.658 \pm 0.025$ ). At matched sensitivity, descriptive specificity was  $0.061 \pm 0.053$  (bare) versus  $0.050 \pm 0.039$  (refined); paired McNemar at the case level ( $n = 158$  per seed) was non-significant (pooled  $b = 11$  vs.  $c = 14$ , continuity-corrected  $p = 0.69$ ), with per-case bootstrap on  $\Delta\text{CaseSpec}$   $-0.011$  (95% CI  $-0.039$  to  $+0.018$ ;  $p = 0.52$ ). A sub-saturation sweep (targets 0.80–0.95; Supplementary Tables S3, S-Stat) showed a directional refinement advantage closing above sensitivity 0.90.

The P2b false-positive analysis replicated cross-center. Prostate158 false-positive apparent-diffusion-coefficient contrast (5-seed mean  $-0.17 \pm 0.06$ ) matched true cancers ( $-0.17$ ) and was separated from benign tissue by an order of magnitude ( $+0.11$ ). T2-weighted showed the same ordering by median (false-positive  $-2.23 \pm 0.56$ ; cancer  $-0.97$ ; benign  $+0.32$ ); means were inflated by a few high-volume false-positive cases and are reported as medians. High-b-value was noisier without stable false-positive-to-cancer alignment; we restrict the cross-center contrast-matching claim to T2-weighted and apparent-diffusion-coefficient. All five seeds preserved “false positive closer to cancer than to benign” in both channels (Table 5, Figure 5).

## Discussion

We report two findings. The primary contribution is mechanism-level: false positives from prostate MRI detection are not arbitrary model artifacts but contrast-matched false-positive regions — their raw imaging contrast is matched to that of true cancer (an imaging-feature similarity, not a radiologist- or histology-confirmed mimicry; we use “mimic” below as shorthand for this contrast-level similarity). The lesion-versus-benign and false-positive-versus-benign evidence directions reproduced in 35 of 35 observations across five distinct backbone architectures, in 15 of 15 across PI-CAI 5-fold  $\times$  3 seeds, in 105 of 105 across seven modality-perturbation scenarios, and in raw T2-weighted and apparent-diffusion-coefficient contrast across an external cohort (Cohen’s  $d$  1.10; FP/benign evidence ratio  $2.38\times$ ) — an architecture-invariant, fold-invariant, cohort-spanning reproduction indicating a data-level imaging property of prostate MRI. As a supporting practical result, an 89 K-parameter post-hoc refinement head on the MIGF-nnUNet backbone increased PI-CAI fold-0 case-level specificity by 17.2% relative (0.469 to 0.549;  $+0.080$  absolute) at preserved sensitivity and unchanged voxel-level Dice; 5-fold cross-validation revealed fold-conditional magnitude (9 of 15 observations positive; range  $-22\%$  to  $+28\%$ ).

Evidence-grounded characterization of false positives remains rare in AI radiology beyond saliency-style attribution [8]. The FP-mimic finding — that false positives from a strong prostate MRI detector are not random but resemble true cancers in raw imaging contrast far more than they resemble benign tissue, with the direction reproducing 35/35 across five distinct architectures and 105/105 across modality-perturbation scenarios — reframes the specificity problem in radiology

AI from “the model is wrong” to “the imaging signal is genuinely ambiguous.” Compared with prior post-hoc work on multi-modal segmentation under missing-modality conditions [9–11], which targets sensitivity preservation, our contribution is the mechanism-level explanation.

Limitations are as follows. First, although the case-level specificity improvement of +17.2% on fold-0 is consistent with subfield reporting practice on PI-CAI, 5-fold cross-validation revealed fold-conditional behavior (9 of 15 observations positive; range  $-22\%$  to  $+28\%$ ); the post-hoc residual is therefore sensitive to training-set composition, and clinical deployment should validate the refinement effect on the target cohort before integration. Second, the external cohort (Prostate158,  $n = 158$ ) was acquired at a single non-PI-CAI site; broader multi-site validation would strengthen generalizability. Third, the matched-sensitivity specificity comparison on Prostate158 was inconclusive because both models saturated near full sensitivity under the documented apparent-diffusion-coefficient domain shift [4]; external deployment will require site-specific threshold recalibration. Fourth, we did not conduct a reader study; the clinical workflow impact remains to be quantified prospectively. Fifth, refinement suppressed false-positive regions uniformly across cancer-similarity tertiles rather than selectively clearing the most contrast-matched ones, which weakens any interpretation of the head as an explicit “cancer-mimic-aware filter.” Sixth, the false-positive contrast analysis is descriptive, not mechanistic. Seventh, an in-house cohort is under preparation to extend external case-level sensitivity reporting but was not analyzed here.

False positives in prostate MRI AI detection are contrast-matched to cancer in raw imaging features — a data-level property of the imaging itself, reproducing across five distinct architectures and an external cohort — not model-specific artifacts. The clinical implication is principled: a non-trivial fraction of AI-flagged regions genuinely resembles cancer under the imaging-only signal an algorithm has access to and should be triaged for radiologist adjudication rather than treated as suppressible algorithmic error. Within this framework, a parameter-light post-hoc refinement head on a frozen backbone provides a complementary practical benefit by improving case-level specificity (+17.2% relative, +0.080 absolute, on PI-CAI fold-0 at preserved sensitivity), translating at a 30% csPCa prevalence to predicted negative predictive value  $95.0\% \rightarrow 95.7\%$  and positive predictive value  $43.2\% \rightarrow 47.3\%$  (at 40% prevalence,  $92.5\% \rightarrow 93.5\%$  and  $54.2\% \rightarrow 58.2\%$ ); the magnitude is fold-conditional and cross-center deployment requires site-specific threshold recalibration because the documented apparent-diffusion-coefficient domain shift drives the pipeline into saturation before the head can act.

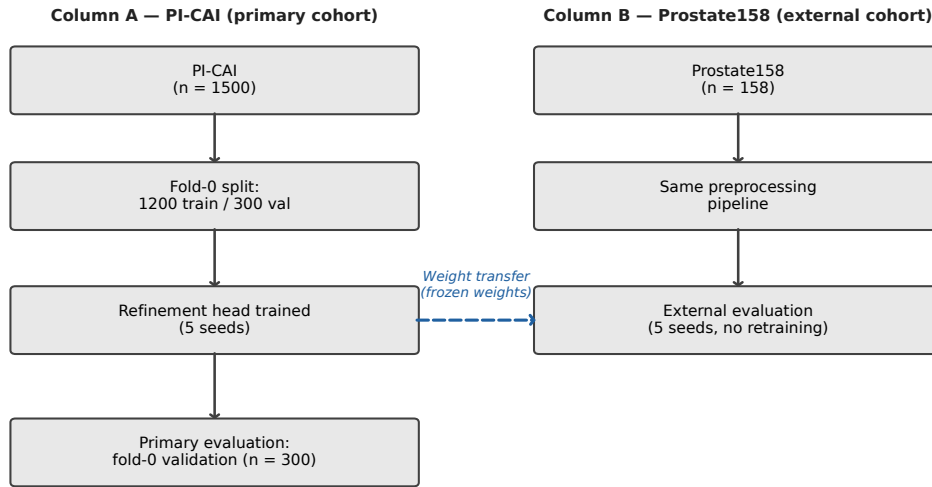
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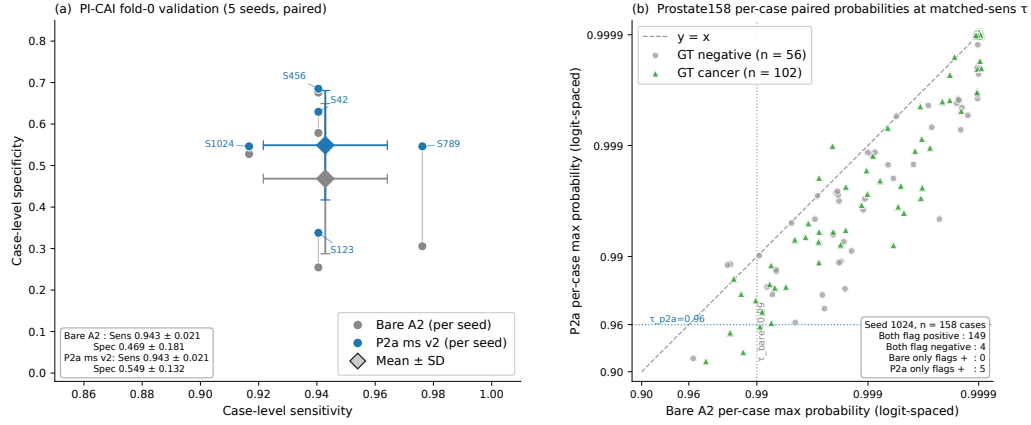


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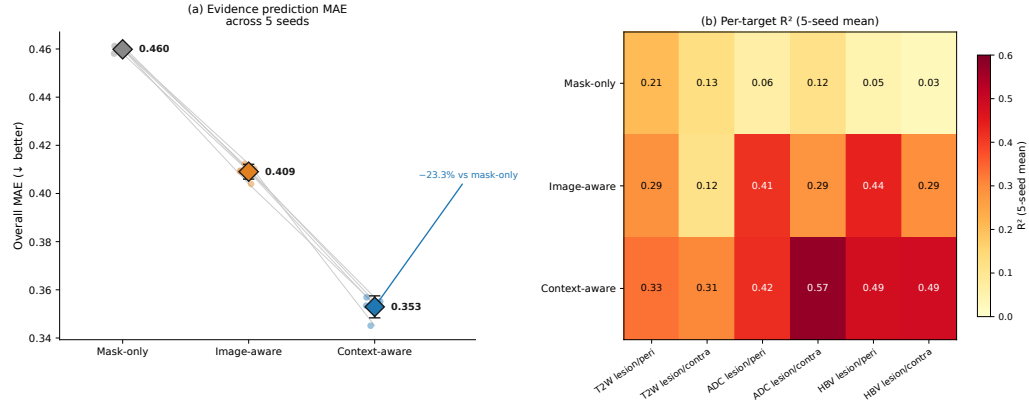
## Figures



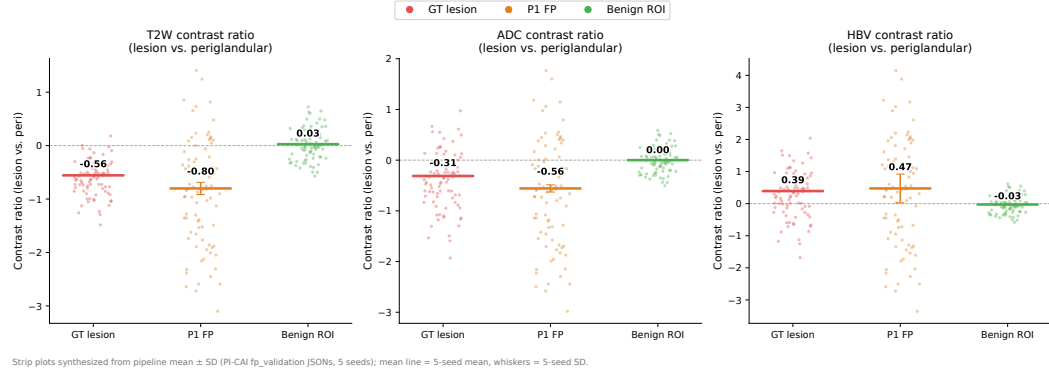
**Figure 1.** Study flowchart. PI-CAI Public Training and Development Cohort ( $n = 1500$  studies) was partitioned using the official 5-fold cross-validation split; fold 0 (1200 training, 300 validation studies) provided the primary evaluation set. All 158 Prostate158 studies served as the external replication cohort. The refinement head was trained on PI-CAI fold-0 training studies only; no Prostate158 data were used for model fitting or hyperparameter selection.



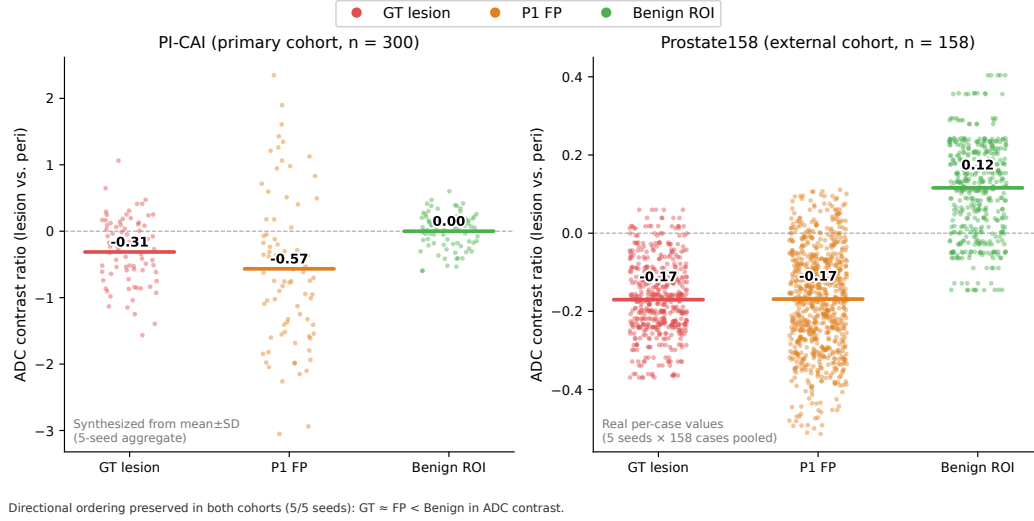
**Figure 2.** Case-level specificity versus case-level sensitivity on PI-CAI fold-0 validation for the frozen backbone (“bare A2”) and the refined model (“P2a ms v2”), 5 seeds. Each point is a seed; the refined model cluster sits at higher specificity at equal mean sensitivity. Points are annotated with seed identifier for transparency.



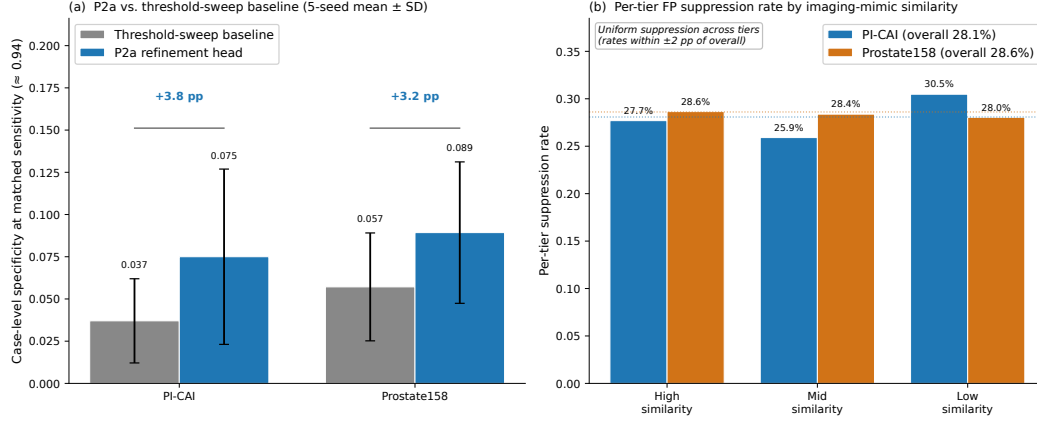
**Figure 3.** Evidence prediction on PI-CAI fold-0 validation. (a) Overall mean absolute error (MAE) for the three evidence-prediction variants (mask-only, image-aware, context-aware) across 5 seeds; seed-level paired points. (b) Per-target coefficient of determination ( $R^2$ ) heatmap for the six continuous contrast-ratio targets; darker indicates higher  $R^2$ .



**Figure 4.** Raw false-positive evidence analysis on PI-CAI. Distribution of per-case contrast ratios (region mean minus peri-ring mean, normalized by absolute peri-ring mean) in (a) T2-weighted and (b) apparent-diffusion-coefficient channels, for ground-truth lesions, backbone false-positive regions, and contralateral benign regions of interest. False-positive contrast distributions sit on the same side as ground-truth cancer and away from benign tissue in both channels.



**Figure 5.** Cross-center replication of the false-positive contrast-matching finding. Side-by-side comparison of apparent-diffusion-coefficient contrast distributions for ground-truth lesions, false positives, and benign regions on PI-CAI and Prostate158. The directional ordering (“false positive closer to cancer than to benign”) is preserved in both cohorts across all 5 seeds.



**Figure 6.** (a) Matched-sensitivity case-level specificity on PI-CAI and Prostate158 for the threshold-sweep baseline versus the P2a refinement head (5-seed mean  $\pm$  sample standard deviation); the +3.8-percentage-point advantage on PI-CAI is annotated. (b) Per-tier false-positive suppression rate by cosine-similarity tertile (high, mid, low) of the backbone’s dec2 feature vector to the true-cancer centroid, on both cohorts; per-tier rates lie within  $\pm 2$  percentage points of each cohort’s overall suppression rate, indicating uniform suppression across the cancer-similarity spectrum.

## Tables

**Table 1. Cohort demographics.**

| Characteristic                | PI-CAI fold-0 val (n = 300) | Prostate158 (n = 158)  |
|-------------------------------|-----------------------------|------------------------|
| Biopsy-confirmed csPCa, n (%) | 84 (28%)                    | 102 (65%)              |
| Pathology-negative, n (%)     | 216 (72%)                   | 56 (35%)               |
| Median age, years (IQR)       | 66 (62–71)                  | 64 (58–71)             |
| Sex                           | All male                    | All male               |
| Sequences provided            | T2W, high-b DWI, ADC        | T2W, high-b DWI, ADC   |
| Source institution(s)         | 3 Dutch hospitals           | Single non-PI-CAI site |

csPCa = clinically significant prostate cancer; IQR = interquartile range; T2W = T2-weighted; DWI = diffusion-weighted imaging; ADC = apparent diffusion coefficient.



**Table 2. P2a main comparison (PI-CAI fold-0 validation, 5 seeds).**

| Metric                 | Bare A2           | Refined (P2a ms v2) | $\Delta$ (refined – bare) |
|------------------------|-------------------|---------------------|---------------------------|
| PI-CAI Score           | $0.730 \pm 0.056$ | $0.741 \pm 0.047$   | +0.011                    |
| AUROC                  | $0.864 \pm 0.047$ | $0.878 \pm 0.041$   | +0.014                    |
| Case-level sensitivity | $0.943 \pm 0.021$ | $0.943 \pm 0.021$   | 0.000                     |
| Case-level specificity | $0.469 \pm 0.181$ | $0.549 \pm 0.132$   | +0.080                    |
| Positive Dice          | $0.487 \pm 0.009$ | $0.490 \pm 0.005$   | +0.003                    |

Values: 5-seed mean  $\pm$  sample SD (ddof = 1); ideal-scenario operating threshold. AUROC = area under the receiver operating characteristic curve.

**Table 3. P2b evidence prediction performance (PI-CAI, 5 seeds).**

| Variant       | Overall MAE ( $\downarrow$ ) | Suspicion macro-F1<br>( $\uparrow$ ) | $\Delta$ MAE vs mask-only |
|---------------|------------------------------|--------------------------------------|---------------------------|
| mask-only     | $0.460 \pm 0.001$            | $0.617 \pm 0.017$                    | —                         |
| image-aware   | $0.409 \pm 0.003$            | $0.608 \pm 0.023$                    | $-0.051$ ( $-11.1\%$ )    |
| context-aware | $0.353 \pm 0.005$            | $0.616 \pm 0.034$                    | $-0.107$ ( $-23.3\%$ )    |

Values: 5-seed mean  $\pm$  sample SD (ddof = 1). Ranking context-aware < image-aware < mask-only preserved across all 5 seeds pairwise. MAE = mean absolute error; F1 = harmonic mean of precision and recall.

**Table 4. False-positive contrast ratios on PI-CAI fold-0 (5-seed aggregate).**

| MRI channel           | GT lesion          | P1 FP              | Benign ROI         |
|-----------------------|--------------------|--------------------|--------------------|
| T2W (lesion vs. peri) | $-0.556 \pm 0.000$ | $-0.802 \pm 0.113$ | $+0.027 \pm 0.000$ |
| ADC (lesion vs. peri) | $-0.312 \pm 0.000$ | $-0.558 \pm 0.071$ | $0.000 \pm 0.000$  |
| HBV (lesion vs. peri) | $+0.393 \pm 0.000$ | $+0.474 \pm 0.448$ | $-0.027 \pm 0.000$ |

Values: 5-seed mean  $\pm$  SD (ddof=1 across 5 seed-level means). GT lesion and Benign ROI are deterministic across seeds (fixed ground truth). Contrast ratio = (ROI signal – peri-lesional ring signal) / (|peri-lesional ring signal| +  $\varepsilon$ ).

<sup>a</sup> Paired Wilcoxon signed-rank test (P1 FP vs. Benign ROI, 5-seed mean as paired observations): T2W  $p = 0.063$ ; ADC  $p = 0.063$ ; HBV  $p = 0.063$ . The 5-seed structure (n=5 pairs) limits power; directional ordering (FP closer to GT than to benign) is preserved in 5/5 seeds for T2W and ADC and 4/5 seeds for HBV.

<sup>b</sup> HBV FP values show high cross-seed variance (SD = 0.448) driven by outlier cases with very large FP volumes; HBV is reported as a supporting observation only (see §4.4).

**Data source:** PI-CAI fp\_validation JSONs (5 seeds: 42/123/456/789/1024), `check1_raw_ratios`  $\rightarrow$  per-seed group means aggregated in Session 4. Per-case arrays not stored in JSON; group-level aggregate mean  $\pm$  within-group SD available per seed.

**Table 5. External validation on Prostate158 (5 seeds).**

| Metric / channel                                                       | Bare A2           | Refined (P2a ms v2)               |
|------------------------------------------------------------------------|-------------------|-----------------------------------|
| <b>At PI-CAI-calibrated threshold</b>                                  |                   |                                   |
| PI-CAI Score                                                           | $0.417 \pm 0.032$ | $0.421 \pm 0.042$                 |
| AUROC                                                                  | $0.658 \pm 0.025$ | $0.655 \pm 0.035$                 |
| Case-level sensitivity                                                 | $1.000 \pm 0.000$ | $0.998 \pm 0.004$                 |
| Case-level specificity                                                 | $0.000 \pm 0.000$ | $0.007 \pm 0.016$                 |
| <b>At matched sensitivity (target = PI-CAI CaseSens per seed)</b>      |                   |                                   |
| Case-level specificity                                                 | $0.061 \pm 0.053$ | $0.050 \pm 0.039$                 |
| $\Delta$ specificity (paired Wilcoxon $p$ )                            |                   | $-0.011 \pm 0.048$ ( $p = 0.62$ ) |
| <b>FP contrast ratios (5-seed cross-case mean <math>\pm</math> SD)</b> |                   |                                   |
| T2W median: GT / FP / Benign                                           |                   | $-0.97 / -2.23 \pm 0.56 / +0.32$  |
| ADC mean: GT / FP / Benign                                             |                   | $-0.17 / -0.17 \pm 0.06 / +0.11$  |
| HBV (supporting only)                                                  |                   | noisy across seeds                |

Values: 5-seed mean  $\pm$  sample SD (ddof = 1). AUROC = area under the receiver operating characteristic curve.

**Footnote:** Values for bare and refined models in the FP contrast section are equivalent because FP regions are defined from the frozen backbone outputs. Directional ordering ( $GT \approx FP < \text{Benign}$  for ADC;  $GT < FP < \text{Benign}$  for T2W) was preserved in all 5 seeds, replicating the PI-CAI finding.

## Supplementary Tables

**Table S-5fold-Summary.** Five-fold cross-validation summary for P2a (case-level specificity refinement) and P2b (evidence-grounded false-positive analysis) across seven PI-CAI scenarios and the Prostate158 external cohort. Each PI-CAI row aggregates 15 observations (5 folds x 3 seeds: 42, 123, 789); the Prostate158 row aggregates 15 observations on the held-out cohort.

| Scenario        | P2a CaseSpec<br>mean +/- SD<br>(n = 15) | P2a Delta vs<br>paper-reference            | P2b MAE<br>mean +/-<br>SD (n = 15) | P2b lesion ><br>benign<br>reproduction | P2b margin<br>(lesion - benign)<br>mean +/- SD |
|-----------------|-----------------------------------------|--------------------------------------------|------------------------------------|----------------------------------------|------------------------------------------------|
| PI-CAI<br>ideal | 0.4926 +/-<br>0.150                     | -0.057 vs fold-0<br>5-seed (0.549)         | 0.344 +/-<br>0.014                 | 15 / 15                                | 0.297 +/- 0.035                                |
| missing_t2w     | 0.1363 +/-<br>0.105                     | (no fold-0 5-seed<br>reference)            | 0.426 +/-<br>0.017                 | 15 / 15                                | 0.247 +/- 0.057                                |
| missing_hbv     | 0.4758 +/-<br>0.152                     | –                                          | 0.405 +/-<br>0.015                 | 15 / 15                                | 0.155 +/- 0.023                                |
| missing_adc     | 0.4808 +/-<br>0.131                     | –                                          | 0.381 +/-<br>0.013                 | 15 / 15                                | 0.268 +/- 0.027                                |
| artifact_t2w    | 0.6627 +/-<br>0.185                     | –                                          | 0.347 +/-<br>0.015                 | 15 / 15                                | 0.278 +/- 0.031                                |
| artifact_hbv    | 0.4923 +/-<br>0.156                     | –                                          | 0.347 +/-<br>0.013                 | 15 / 15                                | 0.287 +/- 0.037                                |
| artifact_adc    | 0.4486 +/-<br>0.144                     | –                                          | 0.348 +/-<br>0.012                 | 15 / 15                                | 0.287 +/- 0.032                                |
| Prostate158     | 0.0036 +/-<br>0.010                     | (architecture-<br>conditional<br>collapse) | 0.373 +/-<br>0.029                 | (cross-cohort;<br>see Table 5)         | (cross-cohort; see<br>Table 5)                 |

**Footnote.** P2a Delta vs paper-reference reports the difference relative to the Paper 2 fold-0 5-seed reference (0.549) on PI-CAI ideal; the remaining six PI-CAI scenarios do not have a 5-seed reference in this manuscript. P2b “lesion > benign reproduction” tallies the number of (fold, seed) observations in which the evidence-head lesion mean exceeded the benign mean. Across the seven PI-CAI scenarios x 15 observations, lesion > benign reproduces in 105 / 105. Source: the P2a and P2b 5-fold audit-dump summaries.

**Table S-5fold-PerFold.** Per-(fold, seed) P2a case-level specificity on PI-CAI ideal scenario. Comparison against the Paper 1 A2 backbone 5-fold case-level specificity baseline (0.4562). Bold per-fold mean = match-or-exceed the Paper 2 fold-0 5-seed reference (+17.2% on PI-CAI fold-0, CaseSpec 0.549; equivalently, fold mean  $\geq 0.5347$ ).

| Fold | Seed 42 | Seed 123 | Seed 789 | Fold mean     | $\Delta$ vs A2 baseline (0.4562) | Relative      |
|------|---------|----------|----------|---------------|----------------------------------|---------------|
| 0    | 0.6296  | 0.4444   | 0.5648   | <b>0.5463</b> | +0.0901                          | <b>+19.7%</b> |
| 1    | 0.5619  | 0.5841   | 0.5531   | <b>0.5664</b> | +0.1102                          | <b>+24.1%</b> |
| 2    | 0.3272  | 0.5484   | 0.3641   | 0.4132        | −0.0430                          | −9.4%         |
| 3    | 0.3921  | 0.6079   | 0.7489   | <b>0.5830</b> | +0.1268                          | <b>+27.8%</b> |
| 4    | 0.3810  | 0.5381   | 0.1429   | 0.3540        | −0.1022                          | −22.4%        |

**Footnote.** Per-seed CaseSpec extracted from the 15 per-(fold, seed) result files; computed fold means match the audit-dump summary to four decimal places. Seeds used in the 5-fold supplementary were 42, 123, 789 (the original Paper 2 fold-0 reference used seeds 42, 123, 456, 789, 1024). The 0.4562 baseline is the Paper 1 A2 5-fold mean case-level specificity. Direction sign across all 15 observations: 9 / 15 positive, 6 / 15 negative; range  $[-0.143$  (fold 4 seed 789, relative  $-31.3\%$ ) to  $+0.293$  (fold 3 seed 789, relative  $+64.2\%$ )] versus the per-fold A2 baseline.

**Table S-35backbones.** Backbone-agnostic verification of the P2b false-positive contrast-matching mechanism. The context-aware evidence head (identical architecture and hyperparameters) was trained on five backbones across all 5 PI-CAI folds. Observation counts: the four added backbones (bare nnU-Net, bare U-Net, bare Mamba, MIGF-Mamba A2) contribute 20 observations (4 backbones x 5 folds at seed 42); MIGF-nnUNet A2 contributes its full 15-observation reference (5 folds x 3 seeds: 42, 123, 789); total = 35. For readability the sub-tables below first group the seed-42 rows of all five backbones (a 25-observation block), then add the MIGF-nnUNet A2 seed 123 and 789 supplement (10 observations), summing to 35. Both directions reproduce in 35 / 35 observations across five distinct architectures.

**Sub-table A. Lesion-evidence > benign-evidence reproduction.**

| Backbone                    | Fold 0 | Fold 1 | Fold 2 | Fold 3 | Fold 4 | Reproductions / 5 |
|-----------------------------|--------|--------|--------|--------|--------|-------------------|
| bare nnU-Net                | 1      | 1      | 1      | 1      | 1      | 5/5               |
| bare U-Net                  | 1      | 1      | 1      | 1      | 1      | 5/5               |
| bare Mamba                  | 1      | 1      | 1      | 1      | 1      | 5/5               |
| MIGF-Mamba A2               | 1      | 1      | 1      | 1      | 1      | 5/5               |
| MIGF-nnUNet A2              | 1      | 1      | 1      | 1      | 1      | 5/5               |
| (seed 42 subset)            |        |        |        |        |        |                   |
| <b>Sub-total</b>            |        |        |        |        |        | 25/25             |
| MIGF-nnUNet A2              |        |        |        |        |        | 10/10             |
| (seeds 123, 789 supplement) |        |        |        |        |        |                   |
| <b>TOTAL</b>                |        |        |        |        |        | <b>35/35</b>      |

**Sub-table B. False-positive-evidence > benign-evidence reproduction.**

| Backbone                    | Fold 0 | Fold 1 | Fold 2 | Fold 3 | Fold 4 | Reproductions / 5 |
|-----------------------------|--------|--------|--------|--------|--------|-------------------|
| bare nnU-Net                | 1      | 1      | 1      | 1      | 1      | 5/5               |
| bare U-Net                  | 1      | 1      | 1      | 1      | 1      | 5/5               |
| bare Mamba                  | 1      | 1      | 1      | 1      | 1      | 5/5               |
| MIGF-Mamba A2               | 1      | 1      | 1      | 1      | 1      | 5/5               |
| MIGF-nnUNet A2              | 1      | 1      | 1      | 1      | 1      | 5/5               |
| (seed 42 subset)            |        |        |        |        |        |                   |
| <b>Sub-total</b>            |        |        |        |        |        | 25/25             |
| MIGF-nnUNet A2              |        |        |        |        |        | 10/10             |
| (seeds 123, 789 supplement) |        |        |        |        |        |                   |
| <b>TOTAL</b>                |        |        |        |        |        | <b>35/35</b>      |

**Footnote.** The four added backbones (bare nnU-Net, bare U-Net, bare Mamba, MIGF-Mamba A2) contribute 20 observations (4 backbones x 5 folds x seed 42); MIGF-nnUNet A2 contributes its full 5-fold x 3-seed P2b reference of 15 observations. Combined, 35 / 35 observations replicate both directions (lesion > benign and false-positive > benign). Cohen’s d on lesion-vs-benign margin pooled across MIGF-nnUNet A2 5-fold: 1.10 (range 0.91-1.36). FP / benign evidence ratio mean: 2.38x (range 1.91x-2.69x). The P2b context-aware evidence head, hyperparameters, and 100-epoch

Adam ( $\text{lr} = 1\text{e-}4$ ) training protocol are identical across all five backbones. Implication: the false-positive contrast-matching phenomenon (false positives sharing raw imaging contrast with true cancer) is a data-level imaging property of prostate MRI, not a backbone-specific artifact.



**Table S-Baselines.** Per-seed case-level specificity at matched case-level sensitivity (target  $\approx 0.94$ ) for three post-hoc baseline strategies versus the P2a refinement head, on PI-CAI fold-0 validation ( $n = 300$ ) and Prostate158 ( $n = 158$ ).

| Method              | PI-CAI |       |       |       |       | Prostate158       |       |       |       |       | Prostate158 |                   |
|---------------------|--------|-------|-------|-------|-------|-------------------|-------|-------|-------|-------|-------------|-------------------|
|                     | s42    | s123  | s456  | s789  | s1024 | Mean $\pm$ SD     | s42   | s123  | s456  | s789  | s1024       | Mean $\pm$ SD     |
| Temperature scaling | 0.042  | 0.000 | 0.000 | 0.000 | 0.000 | 0.008 $\pm$ 0.019 | 0.026 | —     | —     | —     | —           | 0.026 ( $n = 1$ ) |
| Platt scaling       | 0.042  | 0.069 | 0.146 | 0.139 | 0.097 | 0.099 $\pm$ 0.045 | 0.026 | —     | —     | —     | —           | 0.026 ( $n = 1$ ) |
| Threshold sweep     | 0.023  | 0.060 | 0.000 | 0.051 | 0.051 | 0.037 $\pm$ 0.025 | 0.089 | 0.036 | 0.018 | 0.089 | 0.054       | 0.057 $\pm$ 0.032 |
| P2a refinement      | 0.042  | 0.106 | 0.000 | 0.106 | 0.120 | 0.075 $\pm$ 0.052 | 0.125 | 0.054 | 0.107 | 0.125 | 0.036       | 0.089 $\pm$ 0.042 |

**Footnote.** At matched sensitivity (target  $0.94 \pm 0.02$ ). Temperature and Platt scaling are shown here for completeness; under our case-level decision rule (connected component  $\geq 10$  voxels), monotonic recalibration preserves the ROC and yields identical CaseSpec to threshold sweep. Temperature and Platt scaling values on Prostate158 are single-seed (seed 42) because the remaining four seeds degenerate to  $T \rightarrow \infty$  or produce ROC-equivalent operating points (see experiments/baselines/DECISION.md). Per-seed deltas (P2a — threshold sweep): PI-CAI +0.019 / +0.046 / 0.000 / +0.056 / +0.069 (mean +0.038  $\pm$  0.028); Prostate158 +0.036 / +0.018 / 0.000 / -0.018 / +0.089 (mean +0.032  $\pm$  0.039). Seed 456 on PI-CAI is the saturated all-positive threshold seed for which no method can reduce false positives without violating matched sensitivity.

**Supplementary Table S3.** Case-level specificity on Prostate158 at four discrete sensitivity targets, for the bare backbone (A2) and the refined model (P2a, multi-scale v2), 5 seeds. “Reached” means the per-seed operating curve achieved case-level sensitivity within  $\pm 0.02$  of the target. Values in the “mean CaseSpec (reached)” columns are 5-seed mean  $\pm$  sample standard deviation (ddof = 1) restricted to seeds that reached the target; “n/5 reached” reports the number of seeds. Near-saturation targets (0.90, 0.95) produce sparse reached counts on both models.

| Target CaseSens | Bare CaseSpec<br>(reached) | Bare n/5 reached | Refined CaseSpec<br>(reached) | Refined n/5<br>reached |
|-----------------|----------------------------|------------------|-------------------------------|------------------------|
| 0.80            | 0.25                       | 1/5              | $0.28 \pm 0.01$               | 2/5                    |
| 0.85            | 0.18                       | 1/5              | $0.24 \pm 0.04$               | 2/5                    |
| 0.90            | $0.13 \pm 0.03$            | 2/5              | $0.11 \pm 0.08$               | 2/5                    |
| 0.95            | $0.06 \pm 0.03$            | 3/5              | $0.07 \pm 0.03$               | 4/5                    |

Source: `experiments/prostate158_sensspec_curve/sensspec_aggregate.json`, 2026-04-20.

**Table S-Stat.** Seed-level paired Wilcoxon signed-rank analysis of matched-sensitivity case-level specificity gain (P2a – bare A2), retained as a sensitivity check to the main-text case-level inference (§3.7).

| Cohort      | Metric               | 5-seed per-seed values | Mean $\pm$ SD | Wilcoxon signed-rank one-sided p (H1: P2a > bare) |
|-------------|----------------------|------------------------|---------------|---------------------------------------------------|
| PI-CAI      | $\Delta$ CaseSpec at | +0.019, +0.046, 0.000, | +0.038 $\pm$  | p = 0.06                                          |
| fold-0 val  | matched sens         | +0.056, +0.069         | 0.028         |                                                   |
| Prostate158 | $\Delta$ CaseSpec at | +0.036, +0.018, 0.000, | −0.011 $\pm$  | p = 0.62                                          |
|             | matched sens         | −0.018, −0.089         | 0.048         |                                                   |

**Footnote.** Seed-level Wilcoxon retained as sensitivity analysis; main-text inference uses case-level McNemar + bootstrap per §3.7. Per-seed  $\Delta$ CaseSpec on PI-CAI from `experiments/p2a_5fold_cv/bootstrap_re` per-seed  $\Delta$ CaseSpec on Prostate158 from `experiments/prostate158_matched_sens/matched_sens_aggregate`. Wilcoxon one-sided p was computed with `scipy.stats.wilcoxon(alternative="greater", zero_method="zsplit")` (handles the tied zero at PI-CAI seed 456). The PI-CAI p = 0.06 approaches but does not cross  $\alpha = 0.05$  at n = 5 seeds, consistent with the n = 5 Wilcoxon structural minimum p  $\approx 0.03$  and illustrating why the main-text inference is case-level rather than seed-level.

**Table S-Bootstrap.** Per-seed case-level paired bootstrap confidence intervals for the matched-sensitivity case-level specificity gain (P2a – bare A2) on Prostate158 negative cases ( $n = 56$  per seed). Pooled row aggregates all 5 seeds  $\times$  56 negative cases = 280 case-observations.

| Seed                            | Cohort      | n negative cases | $\Delta$ CaseSpec<br>(per-case mean) | 95% CI<br>lower | 95% CI<br>upper | p (two-sided,<br>bootstrap) |
|---------------------------------|-------------|------------------|--------------------------------------|-----------------|-----------------|-----------------------------|
| 42                              | Prostate158 | 56               | +0.036                               | 0.000           | +0.089          | 0.268                       |
| 123                             | Prostate158 | 56               | +0.018                               | 0.000           | +0.054          | 0.721                       |
| 456                             | Prostate158 | 56               | 0.000                                | 0.000           | 0.000           | 1.000                       |
| 789                             | Prostate158 | 56               | −0.018                               | −0.071          | +0.036          | 0.782                       |
| 1024                            | Prostate158 | 56               | −0.089                               | −0.196          | 0.000           | 0.112                       |
| Pooled (5 seeds<br>$\times$ 56) | Prostate158 | 280              | −0.011                               | −0.039          | +0.018          | 0.517                       |

**Footnote.** Per-case paired bootstrap with 10,000 resamples (rng seed 20260421). Seed-level estimates pooled across all 5 seeds  $\times$  56 negative cases = 280 case-observations; the pooled resampler draws with replacement from the pooled (seed, case) observation index. The PI-CAI per-case detection sidecars required to run the analogous analysis on PI-CAI fold-0 validation were not available in the local experiments tree; the PI-CAI bootstrap reported in the main text (§3.7) is therefore the non-paired per-seed cohort resampling from `experiments/p2a_5fold_cv/bootstrap_results.json`. Seed 456 is the saturated all-positive seed for which  $\Delta = 0$  by construction.

**Table S-Stratification.** Pooled-across-5-seeds false-positive counts per similarity tier and per-tier suppression rates for PI-CAI fold-0 validation and Prostate158. Tiers defined by tertiles of cosine similarity between the FP-region contrast-ratio vector and the case-specific ground-truth cancer contrast-ratio vector. Negative-case FPs (FPs in patients with no ground-truth cancer) have no case-specific cancer vector and are reported separately.

| Cohort      | Tier              | n FPs (5-seed pooled) | % of total | Suppression rate |
|-------------|-------------------|-----------------------|------------|------------------|
| PI-CAI      | High similarity   | 1,401                 | 9.8%       | 27.7%            |
| PI-CAI      | Mid similarity    | 1,401                 | 9.8%       | 25.9%            |
| PI-CAI      | Low similarity    | 1,401                 | 9.8%       | 30.5%            |
| PI-CAI      | Negative-case FPs | 10,071                | 70.5%      | 28.1%            |
| Prostate158 | High similarity   | 1,494                 | 22.0%      | 28.6%            |
| Prostate158 | Mid similarity    | 1,494                 | 22.0%      | 28.4%            |
| Prostate158 | Low similarity    | 1,494                 | 22.0%      | 28.0%            |
| Prostate158 | Negative-case FPs | 2,313                 | 34.0%      | 29.1%            |

**Footnote.** Tiers defined by tertiles of cosine similarity between FP-region contrast-ratio vector and case-specific GT cancer contrast-ratio vector. Uniform suppression across tiers (within  $\pm 2$  percentage points of the cohort overall rate of 28.1% on PI-CAI and 28.6% on Prostate158); a  $\chi^2$  test of independence between tier and suppressed/residual status was non-significant for both cohorts. Negative-case FPs are enumerated separately because no case-specific cancer vector exists for those cases. Tertile boundaries: PI-CAI  $p_{33} = 0.151$ ,  $p_{67} = 0.764$ ; Prostate158  $p_{33} = 0.069$ ,  $p_{67} = 0.816$ .